



Paraphrase Detection and Sonnet Generation with LoRA and DPO

Diya Bhattacharjee Kyle Tianshi Jaagat Prashar

diyab@stanford.edu ktianshi@stanford.edu jaagat@stanford.edu

Motivation

Paraphrase detection software is critical to maintaining the integrity of academic work. Sentiment analysis and sonnet generation tests the model's classification and creative modeling abilities.

Question: Do LoRA and DPO, with varied hyperparameters and triplet construction methods, improve output quality and generalization in GPT-2 style transformers for these three tasks?

Methods

LoRA: Injects low-rank trainable matrices into the model, enabling efficient fine-tuning for paraphrase detection as a task.

$$\max_{\Theta} \sum_{(x,y) \in \mathcal{Z}} \sum_{t=1}^{|y|} \log \left(p_{\Phi_0 + \Delta \Phi(\Theta)}(y_t | x, y_{<t}) \right)$$

DPO: Aligns GPT-2 with human preferences without the use of RLHF by fine-tuning the preference data.

$$\nabla_{\theta} \mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_l) - \hat{r}_{\theta}(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_{\theta} \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_{\theta} \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right]$$

DPO Setup

We constructed DPO preference triplets from the Quora training data using two independent methods. Both methods first build positive and negative partner maps from the labeled pairs. For each anchor sentence, its paraphrase partners (label=1) and non-paraphrase partners (label=0) are indexed separately.

Method 1 (Hard Negatives Only): Selects the rejected example for each anchor by finding the non-paraphrase partner with the highest word overlap, using a minimum overlap threshold of 0.2. Anchors with no suitable natural negative are skipped entirely. This yielded ~30k triplets.

Method 2 (Hard Negatives with Synthetic Corruptions): Follows the same hard-negative mining procedure, but instead of skipping anchors that lack a suitable natural negative, generates a synthetic rejected example by randomly shuffling 40% of the words in the chosen sentence. This yielded ~104k triplets.

Each triplet is formatted using the cloze-style prompt template: **Question 1:** ``{anchor}`` / **Question 2:** ``{chosen or rejected}`` / **Are these questions asking the same thing?**

Experiments

Paraphrase Detection — Fine-tuned GPT-2 (12 layers, 768 hidden dimensions, 12 attention heads) on the Quora dataset using a learning rate of 1e-5, batch size of 64, and 10 epochs on an H100 GPU. Then implemented PEFT and DPO extensions.

Model	Dev Accuracy
Baseline	0.82
With PEFT	0.87
With PEFT and GPT-2-medium	0.891
DPO Method 1, Hard Negatives	0.80
DPO Method 2, Synthetic Corruptions	0.60
SMART (Epoch 1)	0.37

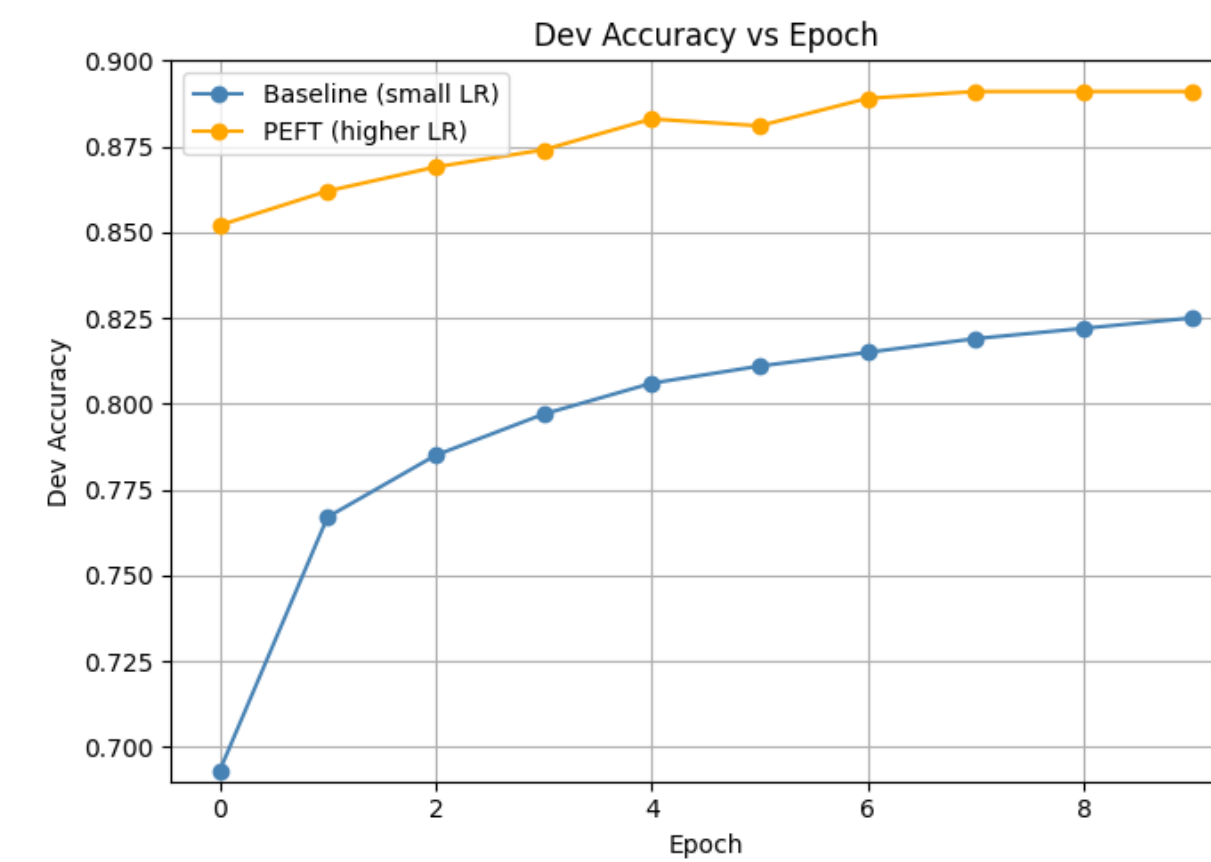


Figure 1. Dev accuracy vs epoch for the baseline (small learning rate) and LoRA with PEFT (higher learning rate). LoRA converges faster and achieves higher final accuracy (0.891) compared to the baseline (0.825).

Sentiment Analysis — Achieved higher accuracy on the CFIMDB dataset than SST.

Model	SST Dev Acc	CFIMDB Dev Acc
Last-Linear-Layer	0.454	0.873
Full-Model	0.497	0.980
Baseline (Last-Linear-Layer)	0.462	0.861
Baseline (Full-Model)	0.513	0.976

Sonnet Generation - Experimented with different learning rates and model sizes. Also incorporated PEFT.

Model	CHRF Score
Baseline (Dev Set)	38.930
With Adjusted LR, Model Size, and LoRA (Dev Set)	42.869
With Adjusted LR, Model Size, and LoRA (Test Set)	41.663

Analysis

Paraphrase Detection: LoRA improved accuracy from 0.82 to 0.87, with GPT-2-medium further improving to 0.891. Neither DPO method reached the baseline of 0.82, likely because DPO was run independently rather than on top of a LoRA fine-tuned model. Method 2 underperformed because the model learned to identify corrupted sentences by their broken syntax rather than by semantic content, which does not generalize to paraphrase detection, where both sentences are grammatically correct. SMART achieved 0.37 after one epoch, likely because it was run from scratch rather than on top of a fine-tuned checkpoint.

Sonnet Generation: GPT-2-medium outperformed GPT-2-small with minimal additional compute cost. Learning rates above 3e-5 overfit the small training set, while rates below 1e-5 did not sufficiently reduce loss. We settled on 3e-5 as our optimal learning rate.

Sentiment Analysis: Full-model fine-tuning outperformed last-linear-layer on both datasets. CFIMDB (0.980) was easier than SST (0.497) due to binary vs. five-category classification.

Conclusion

PEFT via LoRA achieves the greatest gains on paraphrase detection.

DPO does not beat the baseline in its current configuration, with negative example quality being the key limiting factor.

Varying learning rate and model size improved sonnet generation performance.

Future Work

1. Training the paraphrase model on adversarially constructed sentence pairs through the PAWS dataset.
2. Training on 19th-century French Sonnets through the OuPoCo dataset for multilinguality.
3. Experimenting with long-form text for paraphrase detection across paragraphs or pages.

References

- [1] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.
- [2] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. *OpenAI Blog*, 2019.
- [3] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [4] Iona Lacroix and Alexandre Gefen. Oupoco, the oupoco database. *Journal of Open Humanities Data*, 2022.
- [5] Yuan Zhang, Jason Baldridge, and Luheng He. Paws: Paraphrase adversaries from word scrambling. *arXiv preprint arXiv:1904.01130*, 2019.
- [6] Haoming Jiang, Pengcheng He, Weizhu Chen, Xiaodong Liu, Jianfeng Gao, and Tuo Zhao. SMART: Robust and efficient fine-tuning for pre-trained natural language models through principled regularized optimization. *arXiv preprint arXiv:1911.03437*, 2019.